

Trigonometry in 3D



1

a i $EG = 7\sqrt{2}$ cm ii $AG = 7\sqrt{3}$ cm

b i 45° ii 35° iii 90° iv 55°

2

a $z = 14.32$ cm b $\theta = 11.83^\circ$

3

a $z = 21.47$ cm b 75.15 cm²

4

a 11.31° b $CE = \sqrt{325}$ c $BE = \sqrt{334}$ d 9.45°
e $CX = \sqrt{250}$ f $z = 10.74^\circ$

5

a $x = 15$ b $y = 17$ c $\theta = 28^\circ$

6

a i $\alpha = 90^\circ$ ii $\beta = 90^\circ$ iii $\gamma = 90^\circ$
b $x = \sqrt{2}$
c $y = \sqrt{3}$
d $\theta = 35^\circ$

7

a $b = \sqrt{746}$ b 109 cm² c $\theta = 16^\circ$

8

a 91.92 m b 37.14 m c $11^\circ 25'$ d $15^\circ 57'$

9

a $BC = 82$ m b 48 m

10 $\theta = 65^\circ 9'$

11



12

a $MD = \sqrt{32} \text{ cm}$

b $x = 61.87^\circ$

c $h = \sqrt{112} \text{ cm}$

d $MN = 4 \text{ cm}$

e $PN = \sqrt{128} \text{ cm}$

f 69.30°

g 70.53°

13 $AF = 55.77$

14

a $h = 9.37 \text{ m}$

b $\theta = 25^\circ$

15

a $x = 56$

b $h = 90$

16

a $h = 47$

b $x = 131$

c $y = 97$

17

a $\overline{AE} = 26.0 \text{ m}$

b $\theta = 31^\circ 14'$

18

a 2481 m

b 1486 m

c 1384 m

19

a i $VA = 8.49 \text{ cm}$

ii $VC = 7.81 \text{ cm}$

iii $AC = 7.81 \text{ cm}$

b $C = 65^\circ 51'$

20

a $CD = 22.65 \text{ m}$

b $AB = 18.29 \text{ m}$

c 16.18 m

21 41 m

22

a $d = \sqrt{41} \text{ m}$

b $h = 9.86 \text{ m}$

c $x = 63.11^\circ$

d $y = 67.92^\circ$

23

a 1282 m

b 2033 m

c $AB = 1977 \text{ m}$